

Nathan Lee

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EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Mechanical Engineering, 3.61 GPA

Expected May 2028

- Honors & Awards: John Martinson Honors College; Dean's List; Semester Honors; Boeing Engineering Scholarship

WORK EXPERIENCE

Mechanical Engineering Intern

Aug. 2026 – Present

Honda

Russells Point, OH

- Designed and modeled SolidWorks fixtures and assemblies for automated manufacturing applications; developed 3D-printed prototypes to validate fit, function, and manufacturability
- Configured, debugged, and troubleshooted Mitsubishi PLCs supporting automated manufacturing equipment and production processes
- Programmed FANUC 6-axis robotic arms to optimize automated manufacturing processes, with emphasis on motion planning and process efficiency

Undergraduate Teaching Assistant - ENGR 16100/16200

Aug. 2025 – Present

Purdue University College of Engineering

West Lafayette, IN

- Mentor 150+ honors engineering students in engineering problem solving involving mechanics, thermodynamics, data analysis, and computational modeling using MATLAB and Python
- Maintain and troubleshoot laboratory hardware, including Raspberry Pis and sensors, through diagnostics, testing, and software updates
- Evaluate assignments and design projects while providing technical feedback to reinforce fundamental concepts

Product Engineering Intern

May 2026 – Aug. 2026

Metal Components LLC

Grand Rapids, MI

- Modeled SolidWorks CAD assemblies and created 2D manufacturing drawings for sheet metal and tubular components with a focus on design for manufacturability and assembly (DFM/DFA)
- Released Engineering Change Notices (ECNs) to communicate design updates and ensure manufacturing efficiency, documentation control, and production readiness
- Performed manufacturing time studies to identify bottlenecks and reduce process downtime across production

Undergraduate Researcher

Sept. 2025 – Dec. 2025

GetNoisy LLC

West Lafayette, IN

- Evaluated acoustic sensor systems and machine learning-based sound classification algorithms for aircraft noise identification
- Installed, calibrated, and validated acoustic sensors to improve reliability during field deployments
- Applied MATLAB for audio signal processing, feature extraction, classification, and visualization across 1000+ hours of acoustic field data

PROJECTS

Purdue Solar Racing

Jan. 2025 – Present

Vehicle Dynamics Engineer

West Lafayette, IN

- Redesigned the front suspension from a double wishbone to a pushrod configuration, reducing system weight by 20% while optimizing suspension kinematics through calculations and simulations
- Developed and iterated suspension component designs in Siemens NX, using Simcenter NASTRAN FEA across multiple load cases to evaluate stress, deformation, and structural performance
- Implemented ASME Y14.5 GD&T standards to translate suspension designs into detailed manufacturing drawings and managed BOMs for prototype assemblies
- Set up and operated 3-axis and 5-axis CNC machines to machine structural components, including workholding setup and in-process inspection

TECHNICAL SKILLS

CAD: Fusion 360, Siemens NX, SolidWorks

Simulation & Analysis: FEA (Ansys, Siemens Simcenter NASTRAN), Suspension Kinematics

Manufacturing: DFM/DFA, GD&T, CAM, CNC Machining (3-axis & 5-axis), 3D Printing, Laser Cutting

Programming: Python, MATLAB, C